Quiz 1 — Sample 1

DATA 412/612

Weeks 1–4: R and the tidyverse, pipes and functions, ggplot2, and dplyr rows, columns, and groups.

Write your answers in an R script. You may use R and RStudio. You may use your notes and the course website (`https://amousavi1.github.io/612/data-612.html`). You may not use Google, any other website, or an LLM. Load the tidyverse. `mpg` is in `ggplot2`.

1. Write a function `geomean` that takes one numeric vector. Drop every `NA` and every value that is not strictly positive. Using a pipe, return the geometric mean of what remains (`exp` of the mean of the logs). If nothing remains, return `NA`. Try it on `c(1, 4, NA, -2, 0)` and on `c(NA, -1)`.

2. Using `mpg`, make a scatter plot of engine size (`displ`) against highway mileage (`hwy`). Color the points by drive (`drv`) and facet by `year`. Add a title and use `theme_bw()`.

3. From `mpg`, keep cars with 4 or 6 cylinders whose highway mileage is at least 25. Sort by highway mileage, highest first. Return only `manufacturer`, `model`, `cyl`, and `hwy`.

4. For each `class` in `mpg`, report how many cars are in that class and the median highway mileage. Sort by that median, highest first.